

dPVS Demonstration Video

23.8.2002

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This document explains the scenes shown in the dPVS demo video, step by step. The video is roughly six minutes long and divided into six sections, a minute each.

dPVS is an optimizing technology for game engines that enables building and visualizing much more complex environments. The technology is based on dynamic occlusion culling to create the potentially visible set (PVS) at runtime, for every frame.

This demonstration is divided into six parts, showcasing different aspects of the dPVS technology.

The dPVS technology is licensed as a C/C++ library component, available through Criterion Software. There is also a special version of dPVS available which integrates seamlessly with Criterion's RenderWare platform. Inquiries can be sent to the Criterion sales team at sales@csl.com.

The library is available for the PC, Xbox, Sony PlayStation 2 and Nintendo GameCube. This video is captured in real-time from the PC version although all of the versions include the exactly same functionality.

At the writing of this document, dPVS is currently used in Sony's "Star Wars: Galaxies" and "Everquest" series as well as "Tabula Rasa" from NCSoft US and "Project Entropia" from MindArk to mention a few.

The dPVS demonstration from which this video is captured can be downloaded online from <http://www.hybrid.fi/dpvs.html>

The video consists of the following scenes:

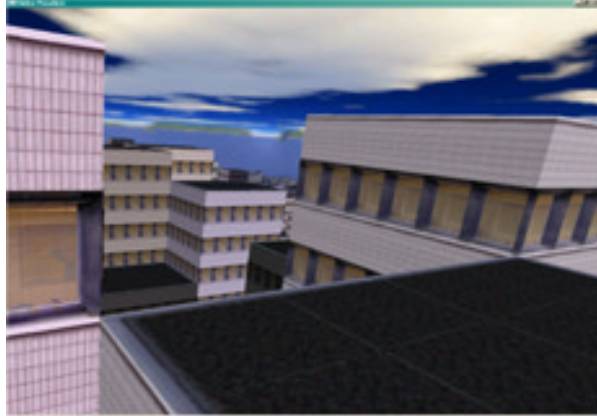
- Scene 1: City Scene
- Scene 2: The Grand Canyon
- Scene 3: Building Scene
- Scene 4: Mirror Scene
- Scene 5: Database Visualization
- Scene 6: The Naked Empire Strikes Back

Any questions regarding the video can be sent to dpvs@hybrid.fi.

Scene 1: City Scene

- 16,000 buildings
- 4,000 moving vehicles

This scene showcases dPVS handling of a Grand Theft Auto style game scenario.



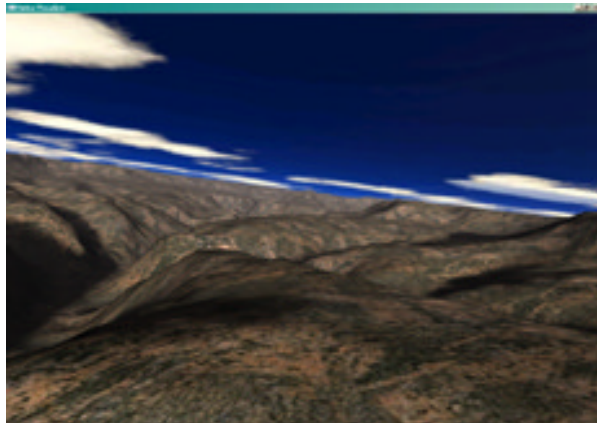
This custom city scene is constructed of 16,000 buildings with 4,000 cars driving around on the streets. It is similar in appearance to many games that take place in urban settings.

When the camera moves near the street level, very little geometry is visible on screen. However, traditionally in 3D graphics everything in the view frustum is included in rendering. DPVS, on the other hand, uses geometry silhouettes to mask out hidden objects. This process is completely dynamic and requires no pre-calculation.

Scene 2: The Grand Canyon

- A 4 million polygon landscape
- Simple LOD.

This scene showcases dPVS handling of landscapes with dynamic changes.



The landscape is created with the type of heightfield models often used in games. The land is divided into rectangular chunks with a simple Level-of-Detail representation.

Each landscape element is a possible occluder object, depending on their estimated contribution to the culling efficiency. When a part of the landscape is lifted from the ground level, that element will start to occlude the parts of landscape hidden behind it.

Scene 3: Building Scene

- Game-like indoor spaces
- Visibility using combination of portals and occlusion

This scene showcases dPVS in a First Person Shooter style of environment and combines pre-determined visibility with dynamic occlusion.



The space is divided into view cells using portals. The portals are manually placed by the artist and tagged as portals for the engine import process.

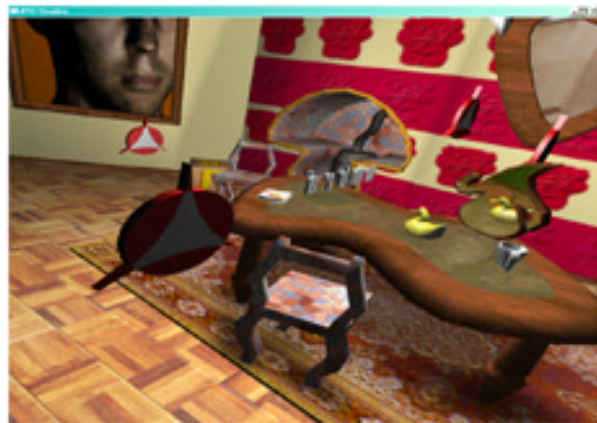
The amount of overdraw in this scene is very low. The combined visibility technologies result in an almost perfect visible set in this scene.

Runtime changes can be introduced by animating the portals or the geometry. Moving objects can hide other objects or even portals to cut off the visibility processing.

Scene 4: Mirror Scene

- Mirrors created using stencils and virtual portals
- Multiple orders of reflection

This scene showcases dPVS ability to maintain high performance even with multiple mirrors and multiple orders of reflection.



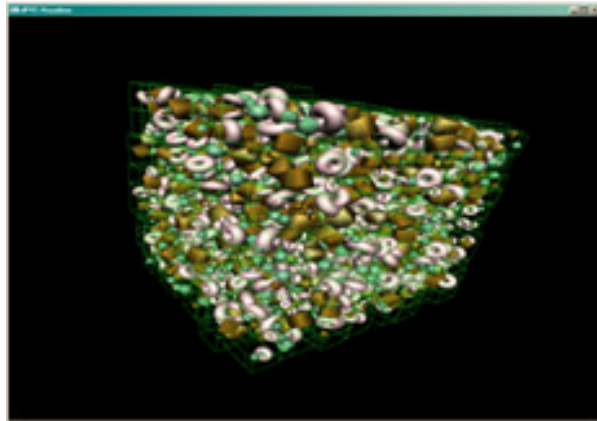
Virtual portals in dPVS can also be used for reflective surfaces. A mirror is effectively just a portal leading into the visibility cell. Both static and moving mirrors can be handled by dPVS. The visibility is resolved consistently to keep mirror performance overhead low.

Mirrors can be built in any shape by using stencil buffering,

Scene 5: Database Visualization

- A Large group of randomly located objects
- A Spatial database built at runtime
- Database refined with moving objects

This scene showcases the adaptive abilities of dPVS in building the whole spatial database without pre-processing.



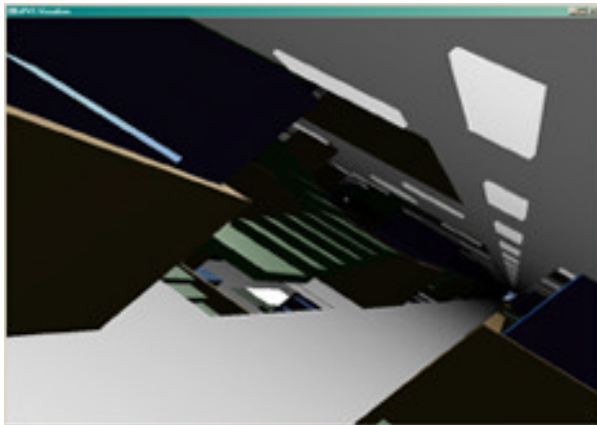
Because the design goal of dPVS was to introduce high-performance occlusion culling to games without preprocessing, even the whole scene spatial organization has to be built at run-time.

The spatial database cell division is visualized in this demo with green lines. When the application launches, dPVS will start to build its internal spatial database, little by little. Each visibility query will further refine the spatial database, adapting and re-balancing the cell division when objects in the scene move.

Scene 6: The Naked Empire Strikes Back

- 280,000 individual objects
- A Complex scene with extremely long lines of sight
- Database refined with moving objects

This scene showcases dPVS in extremely complex environments using the "Naked Empire" model traditionally used in occlusion culling research.



The model was first used by Ned Greene in his seminal occlusion buffer paper published in the Siggraph conference. This version of the model is constructed out of 280,000 separate objects and has complex lines of sight all the way through.

Dynamic occlusion culling based visibility makes it possible to drive scenes of this complexity with real-time performance.

